Week 5 MG.md

THE FINAL EXAM TRAP LIST (Review Before Exam)

- **The "Limiting Nutrient" Trap:** Eutrophication is caused by excess nutrients, but there is a strict rule: **Phosphorus** is the limiting nutrient in freshwater lakes, while **Nitrogen** is the limiting nutrient in marine (ocean) environments.
- **The Carbon Reservoir Trap:** The **Lithosphere (Limestone/Rocks)** is the absolute largest carbon reservoir on Earth. However, if asked what the largest carbon *sink* is, the answer is the **Ocean**.
- **The Decomposition Trap:** Aerobic decomposition produces **stable** products (like $NO_3^-, PO_4^{3-}, SO_4^{2-}$). Anaerobic decomposition produces **unstable**, odorous products (like H_2S, NH_3, CH_4).
- **The Cycle Phase Trap:** The **Phosphorus Cycle** is the only major biogeochemical cycle that does NOT have a gaseous/atmospheric phase.

EVERY Week 5 Concept (Explained Like You're 5)

Part 1: The "Hungry Bugs" Tests (Oxygen Demand)

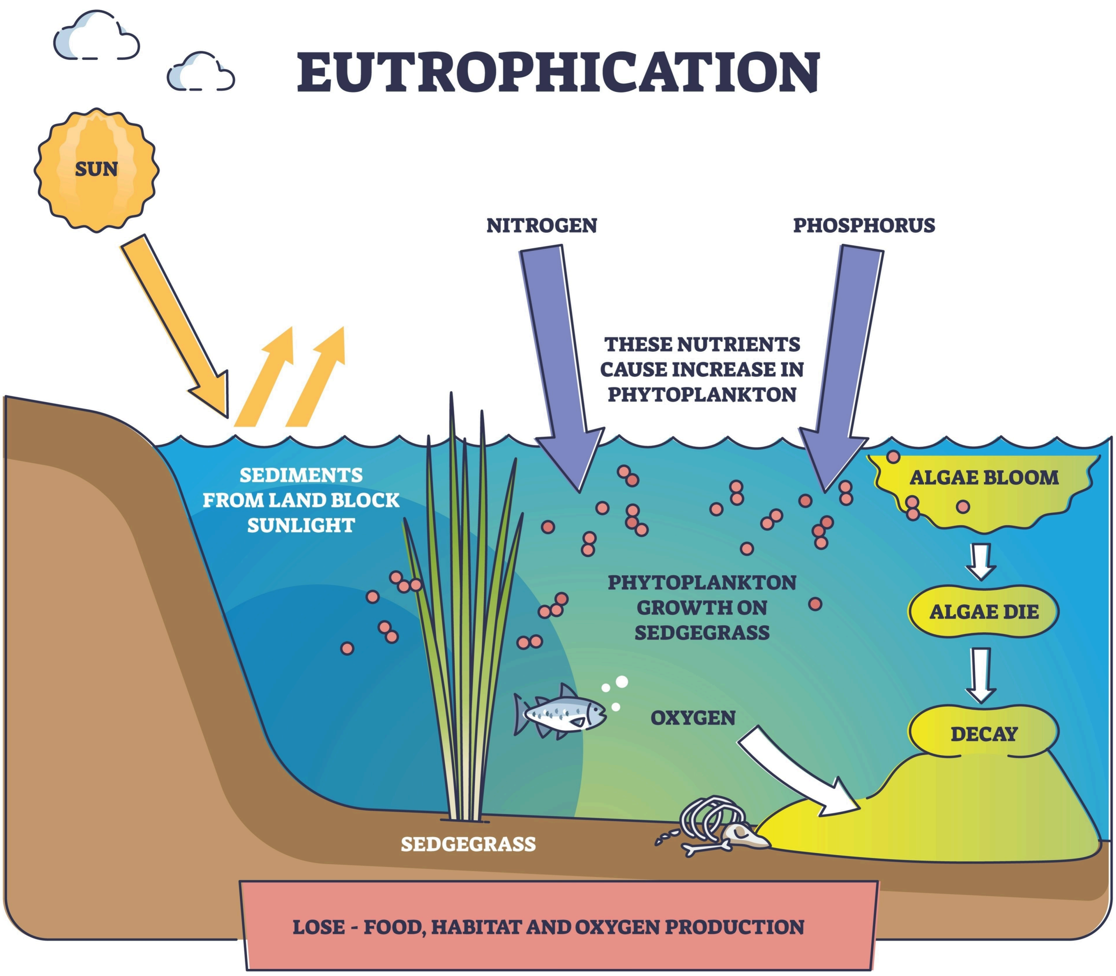
Imagine you throw a sandwich into a fish tank. The invisible bugs (bacteria) in the water will eat it, but they need to "breathe" oxygen to do so.

- **1. BOD (Biochemical Oxygen Demand):** This tests how much oxygen the bugs breathe while eating the biodegradable trash over **5 days at 20°C in the dark**. We "seed" the test (add bugs) to make sure they are present.
  - *The Two Stages:* The bugs eat the easy carbon first (**CBOD**). After 8-10 days, different bugs wake up to eat the nitrogen (**NBOD** / Nitrification).
- **2. COD (Chemical Oxygen Demand):** Instead of waiting 5 days for bugs, we just pour a strong chemical acid (Potassium Dichromate) into the water and boil it at **150°C for 2 hours**. The acid "burns" almost everything—even tough plastics the bugs won't eat.
  - *The Ultimate Rule:* **COD is ALWAYS $\geq$ BOD**.
- **3. ThOD (Theoretical Oxygen Demand):** No bugs, no chemicals. You just use high school chemistry math (stoichiometry) to calculate exactly how much oxygen *should* be needed to destroy a molecule.

- *Rule of Thumb:* **ThOD $\geq$ COD $\geq$ BOD**.

- **Specific Test Conditions & Regulatory Limits (Memorize These):**
    - **Discharge Limits (Surface Water):** BOD must be $\leq 30 \text{ mg/L}$ and COD must be $\leq 250 \text{ mg/L}$.
    - **COD Test Specifics:** The test uses Potassium dichromate ($K_2Cr_2O_7$) as the strong chemical oxidant, and it must be boiled at $150°C$ **for exactly 2 hours**.
    - **Health Limits:** * **Fluoride:** $> 1.5 \text{ ppm}$ causes Fluorosis (bone/teeth damage). $< 0.8 \text{ ppm}$ causes dental cavities.
        - **Nitrate:** $> 45 \text{ mg/L}$ causes Methemoglobinemia (Blue Baby Syndrome).

Part 2: Overfeeding the Lake (Eutrophication)

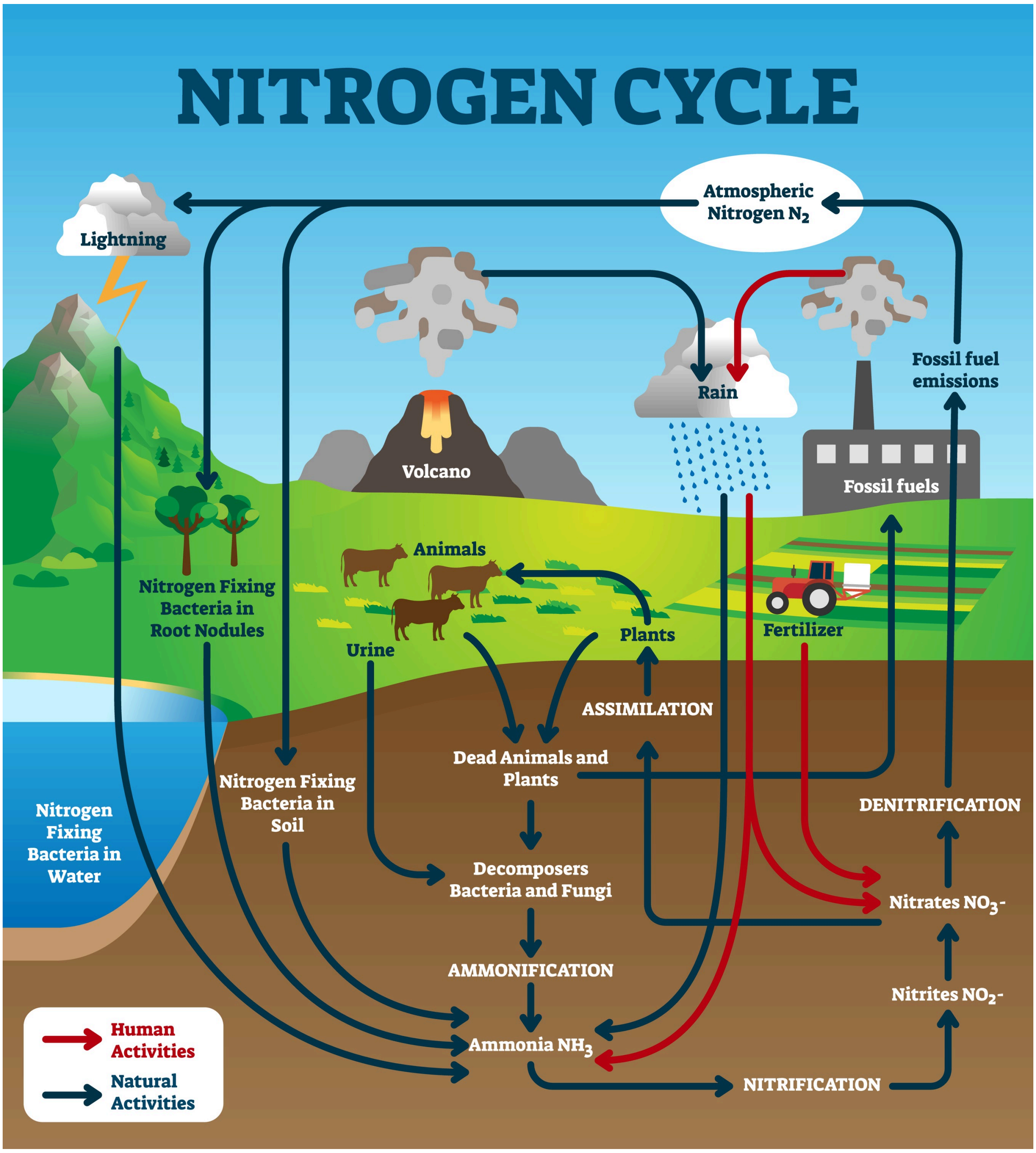


- **4. Eutrophication:** Imagine giving a houseplant way too much fertilizer. It grows like crazy! In a lake, if too much **Phosphorus** washes in from farms, algae explodes in a "bloom." When the algae dies, bugs eat it, suck all the oxygen out of the water, and the fish suffocate.

- *The Trap:* The limiting nutrient in freshwater is **Phosphorus**. In marine/ocean water, it is **Nitrogen**.

Part 3: The Nutrient Cycles (The Planet's Recycling System)

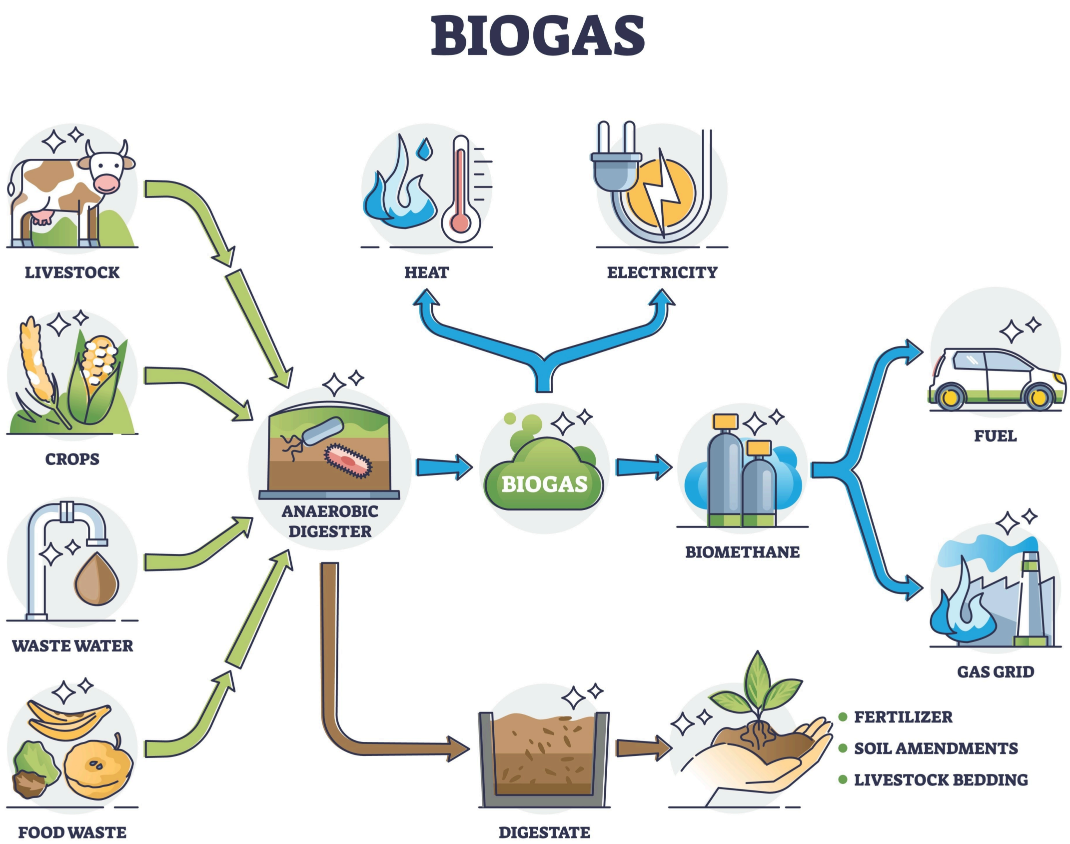
- **5. The Carbon Cycle:** The Earth stores carbon in a giant piggy bank: the **Lithosphere** (rocks/limestone - the largest reservoir). But the **Ocean** is the biggest *active sink* (it is currently absorbing the most CO_2).
- **6. The Nitrogen Cycle (The 5 Steps):**



- Nitrogen gas (N_2) is 78% of the air, but plants can't eat it because it has a super-strong **triple bond**.

    - **Fixation:** Lightning, forest fires, lava, or special bacteria break the bond and pull it into the dirt.
    - **Ammonification:** Dead bugs rot into Ammonia (NH_3). *Free ammonia means very fresh, recent pollution!*
    - **Nitrification:** Ammonia turns into Nitrite (super toxic) $\rightarrow$ Nitrate (safe plant food, indicates *old* pollution).
    - **Assimilation:** Plants eat the nitrate.
    - **Denitrification:** Bacteria turn it back into N_2 gas.
    - *Bonus Fact:* **Kjeldahl Nitrogen** = Organic Nitrogen + Ammonia Nitrogen.
- **7. The Phosphorus Cycle:** The ONLY cycle that has **NO gas phase**. It just moves between rocks, water, and dirt.
- **7.5. The Sulfur Cycle Trap: 99%** of the sulfur in the atmosphere comes from **human activities** (industry), primarily as SO_2, which causes acid rain.
- **Environmental Nitrogen Fixation:** While bacteria do most of the work, nitrogen gas (N_2) is also fixed naturally in the environment by extreme energy events: **Lightning strikes, forest fires, and hot lava flows**.
- **Phytoremediation Specifics:** Remember that phytoremediation strictly involves the use of **plants** (not just bacteria or mechanical filters) to remove, transfer, or stabilize pollutants in soil and sediment.

Part 4: Waste Treatment & Bugs



- **8. Aerobic vs. Anaerobic:**
  - *Aerobic* (with air): Makes stable, safe things like CO_2 and water.
  - *Anaerobic* (without air): Makes unstable, smelly things like H_2S (rotten egg gas), Ammonia, and Methane (CH_4).
- **9. Anaerobic Digestion (The 4 Steps):**
  1. 1 **Hydrolysis:** Chops big chunks into simple sugars.
  2. 2 **Acidogenesis:** Turns sugars into organic acids.
  3. 3 **Acetogenesis:** Turns organic acids into **Acetic Acid**.
  4. 4 **Methanogenesis:** Turns acetic acid into Methane (CH_4) and CO_2 (Biogas for energy!).
- **10. Microbial Fuel Cells (MFCs):** Bacteria eat waste and poop out electrons. If we catch those electrons on a metal plate (anode), we can actually use the bacteria to generate **electricity**.
- **Algae for Fuel:** Among all microorganism groups, **Algae** are specifically highlighted as being used for **biodiesel production**.
- **Phytoremediation:** This is the use of **plants** (not just bacteria!) to remove pollutants (like heavy metals) from soil and water. It includes *phytoextraction* and *phytodegradation*, and is commonly applied in **constructed wetlands**.
- **The Pathogen Match-up Trap:** * *Virus:* Hepatitis A/E, **Rotavirus**
  - *Bacteria:* Cholera, Typhoid

- *Protozoa:* Giardia
- *Helminths:* Worms

The 100% COMPLETE Week 5 Q&A Bank

Oxygen Demand: BOD, COD, ThOD

- **Q:** Why does a high BOD level in a receiving water body pose a threat to aquatic life? **A:** It depletes dissolved oxygen needed by aquatic organisms for survival.
- **Q:** What does a dissolved oxygen (DO) level below the saturation value indicate? **A:** Presence of oxidizable organic matter.
- **Q:** Which statement most accurately distinguishes COD from BOD_5? **A:** COD uses chemical oxidation (including non-biodegradable matter), while BOD relies on microbial oxidation.
- **Q:** In a BOD test, what does the term "seed" refer to? **A:** Microorganisms added to initiate the oxidation process.
- **Q:** A BOD test dilution is considered invalid if: **A:** Final DO falls below 2 mg/L. (And residual DO must be at least 1 mg/L).
- **Q:** In a receiving stream, a high CBOD but negligible NBOD will initially cause which dominant impact? **A:** Dissolved oxygen depletion due to heterotrophic oxidation.
- **Q:** Which oxygen demand term represents the maximum potential oxygen consumption assuming complete biodegradation? **A:** Ultimate BOD ($\mathrm{BOD_u}$).
- **Q:** What is the general relationship between ThOD, COD, and BOD_5? **A:** ThOD $\geq$ COD $\geq$ BOD_5.
- **Q:** What is the discharge limit of BOD and COD in surface water? **A:** BOD = 30 mg/L; COD = 250 mg/L.
- **Q:** Which chemical is used to oxidize organic matter in the COD test? **A:** Potassium dichromate ($\mathrm{K_2Cr_2O_7}$) at 150°C for 2 hours.
- **Q:** Compute the theoretical oxygen demand (ThOD) of water containing glucose ($\mathrm{C_6H_{12}O_6}$) expressed as mg of $\mathrm{O_2}$/mg of glucose.
  - **A:** 1.07.
  - > *Note:* $\mathrm{C_6H_{12}O_6} + 6\mathrm{O_2} \rightarrow 6\mathrm{CO_2} + 6\mathrm{H_2O}$. *The ratio of* $\mathrm{O_2}$ *to glucose is* $(6 \times 32)/180 = 192/180 \approx 1.07$.
- **Q: What are the regulatory discharge limits for BOD and COD into surface water?**
  - **A:** BOD = 30 mg/L; COD = 250 mg/L. [Exam_Ready_Notes]
- **Q:** What is the standard temperature used in the BOD incubation test? **A:** 20°C.
- **Q:** What does a dissolved oxygen (DO) level below the saturation value indicate? **A:** The presence of oxidizable organic matter.

- **Q:** In a BOD test, which measurement is taken immediately after sample preparation (before incubation)? **A:** Initial dissolved oxygen (DO_i).
- **Q:** Why is a dilution step necessary in the BOD test for high-strength wastewaters? **A:** To ensure sufficient dissolved oxygen remains throughout the 5-day incubation.
- **Q:** Which test specifically uses potassium dichromate as the oxidizing agent? **A:** COD test.

Calculations (Use your Formula Sheet!)

- **Q:** 10 mL of wastewater is mixed with dilution water to make 300 mL. We need at least 1.5 mg/L DO drop. What is the minimum detectable BOD_5?
  - **A:** 45 mg/L.
  - > *Note: Dilution factor* $P = 10/300$. *Minimum* BOD_5 = $(1.5)/(10/300) = 45\ mg/L$.
- **Q:** A 2% solution of a sewage sample is incubated for 5 days. DO depletion is 4 ppm. Determine the BOD.
  - **A:** 200 mg/L.
  - > *Note:* $BOD_5 = (DO_i - DO_f)/P = 4/0.02 = 200\ mg/L$.
- **Q:** A wastewater contains 150 mg/L of $C(H_2O)$ and 25 mg/L of $NH_3–N$. What is the total ThOD?
  - **A:** 274 mg/L.
  - > *Note: Carbon demand:* $150 \times (32/30) = 160$. *Nitrogen demand (* $NH_3 + 2O_2 \rightarrow HNO_3 + H_2O$*): requires 4.57 mg* O_2 *per mg N.* $25 \times 4.57 = 114.25$. *Total* = $160 + 114.25 \approx 274\ mg/L$.
- **Q:** How do ThOD and BOD differ in principle? **A:** ThOD is calculated stoichiometrically for complete oxidation; BOD uses microbial oxidation.
- **Q:** A 1% dilution of wastewater shows a DO depletion of 3 mg/L over 5 days. What is the BOD_5? **A:** 300 mg/L.

Eutrophication & Waterborne Diseases

- **Q:** Excessive algal growth when water is overly enriched with minerals and nutrients is called: **A:** Eutrophication.
- **Q:** In the context of eutrophication, which nutrient is typically considered the limiting factor in freshwater ecosystems? **A:** Phosphorus. (Nitrogen is limiting in marine ecosystems).
- **Q:** Which is generally used as a fecal contamination indicator? **A:** *E. coli*.
- **Q:** Which waterborne disease is correctly matched with its causative agent type? **A:** Giardiasis - Protozoa. (Typhoid/Cholera = Bacteria; Hepatitis A = Viral).

- **Q:** “Nitrogen assimilation is the process of formation of organic nitrogen compounds from inorganic nitrogen compounds.”
  - **A:** True.
  - > *Note: Plants take up inorganic nitrate/ammonia and build it into organic DNA and proteins.*

Biogeochemical Cycles

- **Q:** What are the basic requirements for a healthy environment?
  - **A:** Both clean air/water/soil AND safe and adequate food.
- **Q:** Which of the following is the biggest sink of carbon dioxide? **A:** Ocean. (Though Lithosphere/Limestone is the largest *reservoir* overall).
- **Q:** Why is atmospheric N_2 biologically unavailable to most organisms? **A:** Strong triple bond between nitrogen atoms.
- **Q:** Which of the following environmental events fix nitrogen present in air? **A:** Both lightning and forest fires (also hot lava).
- **Q:** Which process converts organic nitrogen from dead organisms back into ammonium (NH_4^+)? **A:** Ammonification.
- **Q:** Which process in the nitrogen cycle converts nitrates (NO_3^-) back into atmospheric nitrogen gas (N_2)? **A:** Denitrification.
- **Q:** Free ammonia in water indicates: **A:** The very first stage of decomposition of organic matter (recent pollution).
- **Q:** Nitrates in water indicate: **A:** Fully oxidized organic matter (old pollution).
- **Q:** Why is the sulfur cycle considered significant for geological processes? **A:** Sulfur drives the formation of minerals in rocks.
- **Q: What percentage of sulfur in the atmosphere is attributed to human activities (industry)?**
  - **A:** 99%. [Notes]
- **Q:** Which biogeochemical cycle does NOT include a gaseous phase? **A:** Phosphorus cycle.
- **Q:** What is the difference between Ammonification and Denitrification? **A:** Ammonification converts organic nitrogen to ammonia; Denitrification converts nitrates back into atmospheric N_2 gas.
- **Q:** Phosphorus cycling primarily involves which three spheres? **A:** Lithosphere, hydrosphere, biosphere.
- **Q:** In the carbon cycle, atmospheric carbon primarily exists in the form of what? **A:** Carbon dioxide (CO_2).
- **Q:** Which process releases stored carbon from the biosphere back into the atmosphere? **A:** Decomposition and respiration.
- **Q:** The primary reservoir of phosphorus in nature is what? **A:** Rocks and minerals (lithosphere).

- **Q:** Which two gases are the major atmospheric components of the carbon cycle? **A:** CO_2 and CH_4.

Anaerobic Digestion & Biotechnology

- **Q:** Which biological treatment simultaneously achieves waste stabilization and energy recovery? **A:** Anaerobic digestion.
- **Q:** Which of the following products is NOT the outcome of aerobic decomposition? **A:** Methane (CH_4). (Methane is produced anaerobically).
- **Q:** What are the end products of the acetogenesis step in anaerobic digestion? **A:** Acetic acid.
- **Q:** Which stage of anaerobic digestion produces methane (CH_4) and CO_2? **A:** Methanogenesis.
- **Q:** Which waste treatment method uses bacteria to drive an electric current? **A:** Microbial fuel cells (MFCs). (Organic matter is oxidized at the anode).
- **Q: Which group of microorganisms is specifically utilized for the production of biodiesel?** * **A:** Algae. [Slide Veto / Notes]
- **Q: Match the Pathogen to its Category (Crucial for Matching MCQs):**
  - **Virus:** Hepatitis A/E, **Rotavirus**.
  - **Bacteria:** Cholera, Typhoid, *E. coli* diarrhea.
  - **Protozoa:** Giardia, Amoebiasis (*Entamoeba histolytica*).
  - **Helminths:** Parasitic worms. [Slide Veto / Notes]
- **Q: What is Phytoremediation?**
  - **A:** The use of **plants** (not just bacteria) to remove, transfer, stabilize, or destroy pollutants in soil and sediment.
- **Q: Define the sub-types of Phytoremediation (Common MCQ Mix-up):**
  - **Phytoextraction:** Plants pull metals into their stems/leaves (e.g., Sunflower).
  - **Phytovolatilization:** Plants soak up pollutants and "sweat" them out as gas (e.g., Weeping willow).
  - **Phytodegradation:** Plants use enzymes to break down pollutants.
  - **Rhizodegradation:** Microbes living around the plant roots break down pollutants. [Notes]
- **Q:** Why does a high BOD level in a receiving water body pose a threat to aquatic life?
  - **A:** It depletes dissolved oxygen needed by aquatic organisms for survival. (The microbes rapidly consume the oxygen while eating the organic matter).
- **Q:** Which biogeochemical cycle does NOT have a gaseous atmospheric phase?
  - **A:** The Phosphorus Cycle.
- **Q:** Environmental nitrogen fixation can occur through which natural events?
  - **A:** Lightning, forest fires, and hot lava flows.

- **Q:** What is the maximum acceptable limit of Nitrate in drinking water, and what disease does it cause if exceeded?
    - **A:** 45 mg/L; it causes Methemoglobinemia (Blue Baby Syndrome).
- **Q:** Which group of microorganisms is specifically highlighted as being utilized for the production of biodiesel?
    - **A:** Algae.
- **Q:** Methane (CH_4) is produced during which type of decomposition process? **A:** Anaerobic decomposition.
- **Q:** Which organisms are responsible for biological nitrogen fixation in the soil? **A:** Nitrogen-fixing bacteria (e.g., Rhizobium).
- **Q:** Which stage of anaerobic digestion produces simple sugars from complex polymers? **A:** Hydrolysis.
- **Q:** Which stage of anaerobic digestion converts simple sugars and amino acids into organic acids? **A:** Acidogenesis.
- **Q:** Which stage of anaerobic digestion produces methane (CH_4) and CO_2? **A:** Methanogenesis.
- **Q:** What is the correct chronological sequence of stages in anaerobic digestion? **A:** Hydrolysis Acidogenesis Acetogenesis Methanogenesis.
- **Q:** Which process uses aerobic microorganisms to decompose organic solid waste into a stable, humus-like material? **A:** Composting.
- **Q:** Which technology uses microorganisms to degrade pollutants in contaminated soil or groundwater? **A:** Bioremediation.
- **Q:** Anaerobic digestion and composting differ primarily in what aspect? **A:** The presence or absence of oxygen (aerobic vs. anaerobic).

ACTIVE RECALL & FEYNMAN CHALLENGE

Cover the answers below and test yourself before the exam.

Quiz 1: Rank BOD, COD, and ThOD in order from highest oxygen demand to lowest. Why are they ranked this way?

> ***Answer:*** *ThOD $\geq$ COD $\geq$ BOD. ThOD is the absolute mathematical maximum based on chemical formulas. COD is slightly lower because the acid might not burn 100% of the toughest molecules. BOD is the lowest because bacteria are picky eaters and will only consume the biodegradable fraction over 5 days.*

Quiz 2: If you are testing a wastewater sample for COD, what specific chemical, temperature, and time duration must you use?

> ***Answer:*** *Potassium dichromate ($K_2Cr_2O_7$) at $150°C$ for 2 hours.*

Feynman Challenge: Explain “Eutrophication” and the concept of a “Limiting Nutrient” to a 10-year-old.

> ***Example Answer:*** *Imagine a lake is like a giant bowl of soup, and the algae are hungry kids. They have plenty of water and sunlight, but they can’t grow because they are missing just one ingredient: Phosphorus (the limiting nutrient). When humans dump fertilizer or poop into the lake, it’s like suddenly dropping a truckload of Phosphorus into the soup. The algae eat it, grow out of control, and cover the top of the lake. When they die, the cleanup crew (bacteria) breathes in all the oxygen in the water to eat the dead algae, leaving no oxygen for the fish, so the fish suffocate.*